

It goes back to the question of how data is perceived by companies and average folk. When the going gets tough, using normal systems and topologies like frameworks just don't cut it. So instead of using a simple linear sequence of layers, functional APIs are used in pretty much everything from neural networks to deep end image processing and text analytical programs. One of the best platforms for building APIs with the functional touch is Keras.

What makes it so great though? No longer do you have to train the models by simply stacking layers to get good accuracies or ROC plots. Built under the Python and pandas library, Keras allows for working on MIMO(multiple inputs multiple outputs), MISO(Multiple Input Single Output) systems. And what about shared layers between different datasets? Functional APIs can deal with such high-level complexities as well.

So take the standard example of a normal classification or regression technique. A functional API may not be the cure-all solution for all kinds of data, but it is the next logical step. A normal sequential API becomes tedious to create because the model has to be built layer by layer. Sharing layers adds flexibility to the models which give birth to a whole breed of data solving methodologies like Siamese and residual networks.

WHY FUNCTIONAL APIs MATTER

Every once in a while you're likely to come across datasets that aren't as simple as the ones you've tested out in tutorials. They may have incomplete or erroneous records with confusing relationships and tough to decipher variable importance.

Take the example of a simple air pollution control dashboard that measures concentration values. The model would pick up on data for temperature, pressure and other topographic information. When the time comes to compile and build a model from it, you'd be hit with a head-scratching issue that infests all datasets-non linearities.

Just drawing a simple correlation plot from the API dashboard might show you that many variables don't have an exact linear relationship with the output that can throw off coefficients through the charts. What makes this process even more challenging is when the API feeds directly on real-time data being generated every moment and can lose track of model accuracies due to disturbances.

While even the fanciest of simple API models can help generate reasonable accuracies, the pitfall often observed is that they fail to work on new